

CHALLIS, ID, USA

WMO: 722142

Lat: 44.523N

Lon: 114.215W

Elev: 1546

StdP: 84.08

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 04114

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-19.0	-26.0	0.4	-21.7	-22.9	0.6	-14.5	7.8	0.7	6.8	0.8	0.4	320	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	33.3	14.3	32.1	13.9	30.2	13.6	16.2	28.0	15.4	27.4	14.6	26.8	4.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	10.9	18.9	11.2	10.0	18.3	9.8	9.1	17.7	50.8	27.9	48.0	26.9	45.5	27.1	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.5	6.3	-26.4	36.2	3.7	1.4	-29.1	37.2	-31.2	38.0	-33.3	38.8	-35.9	39.9		
			-26.6	18.3	3.5	0.9	-29.1	18.9	-31.1	19.5	-33.1	20.0	-35.6	20.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-6.8	-2.5	2.7	6.5	11.6	16.2	21.1	19.4	14.6	7.0	-0.2	-6.7
	DBStd	10.37	6.11	5.28	3.85	3.90	3.92	3.91	2.61	2.61	3.82	4.09	5.10	5.91
	HDD10.0	2187	520	349	226	116	29	3	0	0	9	109	306	519
	HDD18.3	4325	778	583	484	354	211	84	8	19	120	352	555	778
	CDD10.0	1078	0	0	1	12	77	190	344	292	146	16	1	0
	CDD18.3	174	0	0	0	0	1	21	93	52	7	0	0	0
	CDH23.3	3499	0	0	0	6	83	521	1563	1034	285	7	0	0
	CDH26.7	1424	0	0	0	0	12	182	736	419	74	1	0	0
Wind	WSAvg	1.9	1.0	1.6	2.1	2.6	2.5	2.5	2.3	2.0	1.7	1.6	1.4	1.0
Precipitation	PrecAvg	197	12	9	13	15	28	29	18	16	18	10	14	14
	PrecMax	304	43	32	42	43	79	97	66	49	58	46	42	94
	PrecMin	121	0	0	1	0	4	0	0	0	0	0	0	0
	PrecStd	46	9	7	9	11	18	21	14	13	18	11	11	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	12.1	18.2	23.8	28.0	33.2	36.1	34.1	31.3	23.8	14.5	8.9
		MCWB	2.7	5.5	7.1	9.7	12.2	14.3	15.3	14.4	13.7	10.5	7.8	4.3
	2%	DB	4.9	9.0	15.1	20.9	25.9	31.1	34.0	32.4	28.8	20.3	12.2	6.1
		MCWB	1.2	3.4	5.7	8.5	11.2	13.6	14.5	13.9	12.5	9.3	5.7	2.5
	5%	DB	2.8	7.0	12.5	17.8	23.2	28.7	32.7	31.1	27.1	17.5	9.4	3.7
		MCWB	0.0	2.2	4.7	6.8	10.4	12.8	14.2	13.7	12.0	8.4	4.5	0.9
	10%	DB	1.2	4.9	10.1	15.0	20.8	26.7	31.2	29.0	24.2	14.8	7.4	1.4
		MCWB	-1.0	1.0	3.7	6.0	9.6	12.3	13.9	13.2	11.1	7.3	3.4	-0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	6.2	8.0	10.5	13.9	16.3	17.5	16.9	15.4	11.9	8.6	4.7
		MCDB	6.6	11.0	16.1	22.5	22.7	28.9	28.7	27.9	25.5	20.3	13.6	8.2
	2%	WB	1.8	4.1	6.5	8.6	12.3	14.9	16.4	15.8	14.0	10.4	6.4	3.0
		MCDB	4.4	7.8	13.1	18.6	22.5	27.1	28.4	27.1	25.1	17.9	10.8	5.4
	5%	WB	0.5	2.7	5.3	7.5	11.1	13.9	15.7	14.9	12.9	9.1	5.0	1.1
		MCDB	2.5	6.0	11.4	16.5	21.3	25.5	28.0	26.5	24.0	16.1	8.9	3.4
	10%	WB	-0.9	1.3	4.1	6.5	10.1	13.0	15.0	14.1	11.8	7.8	3.5	-0.5
		MCDB	1.0	4.5	9.4	14.1	19.3	24.2	27.6	26.2	22.4	14.3	6.9	1.5
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.9	12.8	13.9	14.4	16.4	19.1	19.1	17.7	14.3	11.4	9.9
		MCDBR	11.0	12.5	16.7	19.1	19.1	20.9	21.5	21.9	21.5	18.2	13.8	9.9
		MCWBR	8.0	7.9	9.4	9.3	8.5	8.1	7.7	8.8	9.6	9.4	8.5	7.2
	5% WB	MCDBR	10.1	11.4	15.0	16.8	17.2	18.5	19.0	19.2	19.3	16.5	12.6	9.3
		MCWBR	7.6	7.5	8.7	8.6	7.9	7.5	7.3	8.1	8.7	8.7	8.1	7.1
Clear-Sky Solar Irradiance	taub	0.245	0.254	0.274	0.327	0.322	0.320	0.329	0.332	0.307	0.291	0.263	0.247	
	taud	2.395	2.365	2.423	2.354	2.456	2.504	2.505	2.465	2.542	2.549	2.546	2.449	
	Ebn at Noon	897	949	971	936	947	947	933	920	923	897	875	862	
	Edh at Noon	72	90	99	116	109	104	103	103	88	76	63	63	
All-Sky Solar Radiation	RadAvg	1.79	2.79	4.12	5.25	5.96	6.87	7.15	6.18	4.85	3.17	1.89	1.39	
	RadStd	0.19	0.29	0.30	0.36	0.58	0.55	0.50	0.36	0.40	0.26	0.16	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+79	N/A

Nomenclature: See separate page